

# XR Renovation Previewer

Evaluation of Interactive Prototype 1, tested in the Week 5 studio on Fri 28 Aug 2026

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Prototype, testing plan, raw data and this evaluation at [github.com/kaikenehme/deco7230-xr](https://github.com/kaikenehme/deco7230-xr) · raw data in Appendices A to D

## 01 OBJECTIVE AND VALIDATION METRICS

### CATEGORY: CREATION

Interactive Prototype 1 is a VR living room built in Unity 6 with the XR Interaction Toolkit and driven by controllers. The floor and sofa are **staying**; everything else can change. The concept turns on one hero interaction, the **hold-up loop**: touch a kept surface and a colour sample comes off it, hold that sample against a wall to preview a harmonised paint, twist your wrist to tune it, and let go to commit. A conventional **controller menu** sits alongside it for comparison, carrying a 24-colour Dulux catalogue, floor materials and eight pieces of furniture. Two secondary diegetic controls round the room out: a lamp that cycles warm, cool and daylight when touched, and a scheme save and cycle.

The testing plan set out three assumptions. **A2 was the primary one**, because the whole concept rests on it and it could genuinely fail.

| AIM | ASSUMPTION UNDER TEST                                                                                                         | MEASURES (DATA SHEET ROW) AND WHAT WOULD COUNT AS SUPPORT                                                                                                                                                                                                                                                    |
|-----|-------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| A1  | People understand "pull a sample off what you're keeping, hold it against what you're changing" <i>without being told</i>     | Time to first unprompted commit (3) · facilitator prompts needed, tallied after ~20 s stuck (4) · twist-to-tune found unprompted (5) · furniture grab found unprompted (16). Support = a commit through the hold-up loop inside Task 1 with few or no prompts                                                |
| A2  | Constraining colour options to what harmonises with the kept surfaces <b>helps</b> people decide rather than frustrating them | Samples tried before first commit (6) · confidence 1 to 5 (9) · "did the limited options feel helpful or restrictive?" (10, <b>decisive</b> ) · think-aloud on rejected candidates (11). Support = participants who used the constrained set call it helpful and report confidence not lower than menu users |
| A3  | With both routes available, which one people reach for, which produces the kept choice, and which they trust                  | Method used first after the Task 1 prompt (13) · method that produced the kept choice (14) · post-test "which did you trust more?"                                                                                                                                                                           |

The plan fixed the measures and the 20-second prompt rule but never set numeric pass marks. Rather than invent thresholds after the fact, I judged each aim on the *direction* and *consistency* of the evidence across participants, and I say so openly in §4. Each session ran for about five minutes: a short brief, thirty seconds of uninstructed free look, Task 1 ("*repaint this room so it works with the floor and sofa you're keeping*"), Task 2 (change the floor, add and move furniture, save and switch schemes), then four post-test questions. One paper sheet per participant, sixteen rows; the sheet itself is in Appendix D.

**Setting, read this before the results.** The studio session ran on the **XR Device Simulator** inside the Unity Editor on a laptop, with mouse and keyboard standing in for the headset and controllers (grip = G, trigger = click, wrist twist = mouse-driven controller rotation). The borrowed Quest refused to authorise over USB, so the built APK was never sideloaded. Every "controller" action below was really a mouse action. That confounds every discoverability measure, and I treat it that way throughout.

## 02 RESULTS

The studio session produced **two participants**, P1 and P2, both course peers. The plan aimed for five or more but studio time allowed two, so over the following weekend I ran **three supplementary sessions** at home with my flatmates (P3 to P5), on the same simulator setup, with the same printed sheet and the same protocol. Those three are **out-of-class data**. I report them separately in every table, and counts are always given as studio plus supplementary. No session was screen recorded, since consent was declined or not taken. Full rows and observer notes are in Appendices A to C.

### What participants did

| BEHAVIOUR                                                 | P1          | P2          | P3          | P4          | P5          | COUNT (STUDIO + SUPP.)                 |
|-----------------------------------------------------------|-------------|-------------|-------------|-------------|-------------|----------------------------------------|
| First thing reached for, uninstructed                     | wall        | —           | floor       | sofa        | wall        | kept surface first: 2 of 4 recorded    |
| Touched a kept surface (floor/sofa) in free look          | yes         | yes         | yes         | yes         | yes         | 5/5 (2 + 3)                            |
| Pulled a sample or held one against a wall (hold-up loop) | no          | no          | no          | no          | no          | 0/5 (0 + 0)                            |
| Discovered twist-to-tune                                  | —           | —           | no          | no          | no          | 0/3 recorded (no sample was ever held) |
| Opened the menu unprompted                                | no          | no          | —           | no          | —           | 0/3 recorded                           |
| Method used first · method that produced the kept choice  | menu · menu | menu · menu | menu · menu | menu · menu | menu · menu | menu 5/5 (2 + 3)                       |
| Facilitator prompts needed                                | 5           | 3           | 4           | 2           | 4           | mean 3.6 in a ~5 min session           |
| Samples tried before first commit                         | 3           | 2           | 4           | 2           | 2           | all via the menu                       |
| Time to first unprompted commit                           | —           | ~3 min      | ~2 min      | ~2 min      | ~1 min      | minutes only; seconds not logged       |
| Grabbed furniture (grip) unprompted                       | no          | no          | no          | yes         | no          | 1/5 (0 + 1)                            |
| Used the lamp unprompted                                  | no          | no          | no          | yes         | yes         | 2/5 (0 + 2)                            |
| Saved or cycled schemes (Task 2)                          | neither     | —           | neither     | neither     | neither     | 0/4 recorded                           |

What participants said (verbatim from the sheets)

| QUESTION                                               | P1                                      | P2                                                                                    | P3                                      | P4                                                                    | P5                                                               |
|--------------------------------------------------------|-----------------------------------------|---------------------------------------------------------------------------------------|-----------------------------------------|-----------------------------------------------------------------------|------------------------------------------------------------------|
| Confidence in final choice, 1 to 5                     | 3                                       | —                                                                                     | 3                                       | 4                                                                     | 2                                                                |
| "Did the limited options feel helpful or restrictive?" | "It's good"                             | "Enough for now"                                                                      | "Feels a bit restrictive"               | "Feels good for now"                                                  | "Need more painting options"                                     |
| Why was a candidate rejected?                          | —                                       | —                                                                                     | "The walls highlight"                   | "The furniture drag and drop"                                         | "Furniture selection"                                            |
| One thing that confused you                            | "Simulator is confusing"                | "Drag and drop (hard)"                                                                | "The controller"                        | "The controller"                                                      | "Controller"                                                     |
| Observer notes                                         | "Changing the highlight colour (frame)" | "Toggle to rotate furniture" · "Once you select the furniture you PLACE IT, not drop" | "Change the highlight for a framed one" | "When we select a furniture we should place it, not dropped in front" | "Have more options on furniture and maybe a room already set up" |

**Data quality.** The studio sheets captured 11 and 10 of their 16 rows; the supplementary sheets captured 15 each. Nobody's sheet records which of the six scripted prompts was used, only how many. The post-test question about which method participants trusted more was in the plan but never made it onto the printed sheet, so it was never asked. Row 11, which was meant to capture why a colour was rejected, was answered by all three supplementary participants as a list of what was hard instead. And row 10 was asked, exactly as printed, of five people who had all chosen through the menu, so every answer describes the 24-colour catalogue rather than the constrained set the question was written for.

03 ANALYSIS AND INSIGHTS

Five participants across two settings is a small sample. I report counts rather than percentages, and I only call something a pattern when it held for every participant it applied to.

**The clearest result is that the loop's entry gesture happens and the loop never starts.** All five participants touched the floor or the sofa during free look, which is exactly the gesture that begins the hold-up loop, and none of them went on to pull a sample. Touching a kept surface simply gives no response that reads as "you can take this". The result survived a change of setting, a facilitator who had already run the protocol, and participants who probably knew what the project was, so I trust it as far as five people can be trusted. I read it as a failure of *affordance* rather than of the concept: the interaction was never reached, so it was never judged.

**The menu won every time, but by default rather than by preference.** Both routes to a paint choice were invisible. The menu produced all five kept choices only because the fifth scripted prompt ("point at the floor and press the trigger") reveals it directly, while nothing in the room reveals the hold-up loop. Of the three sheets that record how the menu was reached, all three say it took a prompt. So this is a discoverability result, and I do not read it as people preferring menus.

**Prompting stayed constant** at roughly one intervention per minute, 5 and 3 in the studio and then 4, 2 and 4 at home, against a 20-second stuck rule. Practice and a familiar setting changed nothing, which points at the prototype rather than at the sessions.

**Everything participants found confusing was about manipulation, not choice.** The confusing thing was the controller (three times), drag and drop (twice), the selection highlight (twice), or rotating furniture, and P2 and P4 independently said almost the same sentence: selecting furniture should *place* it, not drop it at your feet. Nobody was confused by the colours, the options or the room. Some of this is simulator cost, a mouse standing in for a hand, but place versus drop and the highlight are prototype decisions and would look the same on a headset.

**The diegetic lamp can be found, just not in the studio.** Two flatmates touched it unprompted; neither studio participant did. The data cannot separate a practised facilitator, prior exposure to the project and a more relaxed free look, so I leave the why open. The scheme save and cycle was found by nobody in either setting.

**Opinions on the limited options split once n grew, and they were about the wrong thing.** The studio pair said "it's good" and "enough for now"; at home I got "feels good for now" but also "feels a bit restrictive" and "need more painting options". All five were rating the unconstrained 24-colour catalogue. That says nothing about A2 directly, but it is a warning: if 24 free colours already feel thin to two people out of five, a seven-option harmony set will read as *fewer* unless the constraint visibly explains itself.

04 EVALUATION OF AIMS

| AIM | VERDICT                                 | GROUNDS                                                                                                                                                                         | HOW MUCH WEIGHT IT CAN CARRY                                                                                                                                    |
|-----|-----------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| A1  | <div>INVALIDATED</div> on the simulator | 0/5 pulled a sample; 0/3 recorded found the menu unprompted; 2 to 5 prompts per person; twist-to-tune never seen                                                                | Moderate. It held across two settings and two participant pools, but the simulator caps the claim at "invalidated with a mouse" rather than "invalidated in VR" |
| A2  | <div>UNTESTED</div> (the primary aim)   | No participant ever held a harmonised sample, so the constrained set was never seen. The row 10 answers (5/5) and the confidence scores (3, 3, 4, 2) all belong to menu choices | None. There is no data on the assumption, and the three extra sessions changed nothing because the gate to A2 never opened                                      |
| A3  | <div>HALF ANSWERED</div>                | Used: menu 5/5, always after prompting where recorded. Trusted: never asked, because the question was not on the printed sheet                                                  | Low. The menu winning is a consequence of A1 failing, not a comparison of two working routes                                                                    |

The honest summary is that IP1 tested the prototype's **discoverability on the simulator** and found the hold-up loop undiscoverable five times out of five. It did **not** test the concept's central claim. I would rather say that plainly than report a soft "A2 looked fine" from answers that were really about a different feature. The negative A1 result is useful precisely because the uninstructed phase exposed it; a guided demo would have hidden it.

## 05 CONCEPT ITERATION

Each change below answers a row in §3. The concept itself (match against what is staying, hold it up, no menus) is **affirmed as untested rather than changed**: nothing in the data argues against it, because nothing in the data reached it. The iteration is aimed at getting participants *to* it.

| INSIGHT (FROM §3)                                                                   | CHANGE, FIX OR AFFIRMATION                                                                                                                                                                                                                                                                                                                        | WHERE             |
|-------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|
| Touching a kept surface does nothing visible, so the loop never starts (5/5)        | Give kept surfaces a <b>first-contact affordance</b> : a persistent pullable tab, a swatch hovering at the touch point, a hand glow on proximity, and a sample that visibly <i>peels</i> off on trigger. Feedback at the moment of contact rather than a tutorial. This is the weakest thing IP1 exposed, and the thing IP2a is meant to rebuild. | IP2a build        |
| Both routes invisible; the one a prompt names wins                                  | Accept that <b>diegetic is not the same as discoverable</b> . On entry the room gets a ten-second in-world onboarding: the floor and sofa pulse once under a "staying" label and the lamp glows. No text panels and no menus, so it stays diegetic without staying silent.                                                                        | IP2a build        |
| Complaints are about manipulation: place vs drop (twice), highlight (twice), rotate | Make grip <b>hold</b> and release <b>place</b> where the ray lands, which is P2 and P4's own suggestion; put rotate on the thumbstick; turn the selection highlight into a clear outline instead of a tinted frame.                                                                                                                               | IP2a build        |
| A2 never reached after five sessions                                                | <b>Change the test rather than the design</b> . IP2a's Task 1 starts with a sample already in the participant's hand, so everyone reaches the constrained set and row 10 finally measures what it was written for. Free discovery of the entry gesture is measured separately in the uninstructed phase.                                          | IP2a test plan    |
| 2/5 called a 24-colour catalogue restrictive                                        | Let the <b>constraint explain itself</b> . Each harmonised option shows why it matches, with a thread of colour running back to the floor or sofa it was derived from. Constraint that is visibly doing something reads as help; constraint that is silent reads as fewer options.                                                                | IP2a build + test |
| Lamp 0/2 studio, 2/3 home; save/cycle 0/4                                           | <b>Keep the lamp</b> , since it can be found, and give it a pull cord and a glow so studio-condition users find it too. Scheme save and cycle was found by nobody, so it either moves to a visible object such as a pinboard on the wall or comes out of the IP2a test.                                                                           | IP2a build        |
| Furniture drew the opinions ("more options", "room already set up")                 | Start IP2a from a <b>furnished room</b> so the task is rearranging rather than furnishing from empty. Furniture grab, move and rotate was already planned for IP2a and is now the second priority after the entry affordance.                                                                                                                     | IP2a build        |

## 06 REFLECTION AND NEXT STEPS

**What worked.** The uninstructed phase did its job. It produced an unambiguous negative on the entry gesture that a guided demo would have hidden, and keeping "did" separate from "said" on the sheet made the pattern in §3 easy to see. Adding three real supplementary sessions was cheap, and it turned a 2 out of 2 observation into a 5 out of 5 one without changing a single verdict, which is itself worth knowing.

**What didn't.** The platform. A body interaction was tested through a mouse, five times, because the Quest would not authorise over USB on the day. The same pipeline risk hangs over IP2b, so it gets solved first rather than last. Facilitating alone with a paper tally lost about a third of the studio rows; at home, with practice, it lost almost none, but the same instrument gaps came back because I never revised the sheet between rounds.

**Instrument.** The IP2a sheet gets three fixes. Row 10 branches, so the constraint question is only asked of people who actually used the constrained set, and menu users are asked why the menu instead. Row 11 is reworded so it stops collecting "what was hard". And the trusted-method question gets printed on the sheet instead of living in the plan. Beyond those, prompt *numbers* rather than ticks go into the tally boxes so I know which prompt unstuck whom, and sessions are screen recorded with consent so lost rows can be recovered.

**Sample.** Flatmates got n to five, but they bring prior exposure, which may explain the lamp result. Next round I recruit outside the flat, log whether the participant has seen the project before, and budget seven minutes per participant rather than five, because setup and briefing ate the studio's time.

**Next.** IP2a runs on the headset, with the entry affordance rebuilt, Task 1 starting sample in hand, and A2, still the primary aim, finally measured.

# Appendices

A studio results · B supplementary results · C observer and transcription notes · D data sheets

## A STUDIO SESSION RESULTS, FRI 28 AUG 2026, WEEK 5 STUDIO (N = 2)

Transcribed from the two paper sheets on Sun 30 Aug. Blank = not recorded. Platform: XR Device Simulator (see §1). Facilitator: Kaike Nehme, solo. No recording.

| #  | MEASURE                                               | P1                                          | P2                                          |
|----|-------------------------------------------------------|---------------------------------------------|---------------------------------------------|
| 1  | Reached for first (free look)                         | wall                                        |                                             |
| 2  | Floor/sofa touched during free look                   | yes                                         | yes                                         |
| 3  | Time to first unprompted commit                       |                                             | 3 m (seconds not recorded)                  |
| 4  | Facilitator prompts needed (tally)                    | 5                                           | 3                                           |
| 5  | Twist-to-tune discovered                              |                                             |                                             |
| 6  | Samples tried before first commit (tally)             | 3                                           | 2                                           |
| 7  | Scheme save / cycle used (Task 2)                     | neither                                     |                                             |
| 8  | Lamp / light discovered or used                       | no                                          | no                                          |
| 9  | Confidence in final choice (1 to 5)                   | 3                                           |                                             |
| 10 | "Limited options: helpful or restrictive?" (verbatim) | "It's good"                                 | "Enough for now"                            |
| 11 | Think-aloud: why was a candidate rejected?            |                                             |                                             |
| 12 | One thing that confused you (verbatim)                | "Simulator is confusing"                    | "Drag and drop (hard)"                      |
| 13 | Method used first after Task 1 prompt                 | menu                                        | menu                                        |
| 14 | Method that produced the kept choice                  | menu                                        | menu                                        |
| 15 | Menu opened unprompted? / furniture placed?           | prompted · (placed/moved/failed not marked) | prompted · (placed/moved/failed not marked) |
| 16 | Furniture grab (grip) discovered                      | prompted                                    | prompted                                    |

The post-test question "which method did you trust more?" was not on the sheet, so no answer was recorded.

## B SUPPLEMENTARY SESSIONS, WEEKEND OF 29 TO 30 AUG 2026, AT HOME (N = 3, OUT OF CLASS)

Three flatmates took part. It is a convenience sample and they had probably heard about the project beforehand. Same simulator setup, same printed sheet, same protocol, same facilitator. The instrument fixes listed in §6 were not applied before these sessions ran. The sheets carry the pre-printed studio date, which should be ignored. Screen-recording consent was declined by all three.

| #  | MEASURE                                               | P3                        | P4                            | P5                           |
|----|-------------------------------------------------------|---------------------------|-------------------------------|------------------------------|
| 1  | Reached for first (free look)                         | floor                     | sofa                          | wall                         |
| 2  | Floor/sofa touched during free look                   | yes                       | yes                           | yes                          |
| 3  | Time to first unprompted commit                       | 2 m                       | 2 m                           | 1 m                          |
| 4  | Facilitator prompts needed (tally)                    | 4                         | 2                             | 4                            |
| 5  | Twist-to-tune discovered                              | no                        | no                            | no                           |
| 6  | Samples tried before first commit (tally)             | 4                         | 2                             | 2                            |
| 7  | Scheme save / cycle used (Task 2)                     | neither                   | neither                       | neither                      |
| 8  | Lamp / light discovered or used                       | no                        | unprompted                    | unprompted                   |
| 9  | Confidence in final choice (1 to 5)                   | 3                         | 4                             | 2                            |
| 10 | "Limited options: helpful or restrictive?" (verbatim) | "Feels a bit restrictive" | "Feels good for now"          | "Need more painting options" |
| 11 | Think-aloud: why was a candidate rejected? (verbatim) | "The walls highlight"     | "The furniture drag and drop" | "Furniture selection"        |
| 12 | One thing that confused you (verbatim)                | "The controller"          | "The controller"              | "Controller"                 |
| 13 | Method used first after Task 1 prompt                 | menu                      | menu                          | menu                         |
| 14 | Method that produced the kept choice                  | menu                      | menu                          | menu                         |
| 15 | Menu opened unprompted? / furniture placed?           | (not marked) · placed     | prompted · (not marked)       | (not marked) · placed        |
| 16 | Furniture grab (grip) discovered                      | prompted                  | unprompted                    | prompted                     |

## C OBSERVER NOTES AND TRANSCRIPTION NOTES

### Observer notes, verbatim

- P1 — "Changing the highlight colour (frame)" · "Time on task [illegible] think [illegible]"

### Transcription notes

- Tallies were read from ticked boxes at high zoom. Which of the six scripted prompts was used was not recorded on any sheet, only the count.

- **P2** — "Toggle to rotate furniture" · "Once you select the furniture you PLACE IT, not drop"
- **P3** — "Change the highlight for a framed one"
- **P4** — "When we select a furniture we should place it, not dropped in front"
- **P5** — "Have more options on furniture and maybe a room already set up"
- P1's "simulator is confusing" and the three "the controller" answers refer to the mouse and keyboard emulated controller of the XR Device Simulator. They are treated as a setting confound, not as a finding about a Quest controller.
- Row 10 was asked of participants who chose via the menu (row 14 = menu, 5/5), so the answers describe the 24-colour catalogue, not the harmony-constrained set.
- Row 11 (P3 to P5) was answered as "what was hard" rather than as a reason for rejecting a colour, and is treated as extra confusion items.
- Row 15 for P3 and P5: only "placed" is ticked, so whether the menu was opened unprompted is not recorded.
- The ink on the P3 to P5 sheets was very light, and two cursive readings are approximate: "framed" (P3's note) and "highlight" (P3, row 11).

## D DATA SHEETS: SCANS OF THE FIVE PAPER SHEETS (CONTRAST-ENHANCED FOR LEGIBILITY)

One sheet per participant, sixteen rows plus the six scripted prompts and observer notes. Originals as PDF in `testing-data/ip1/sheets/`. No identifying information was written on any sheet.

**IP1 data collection sheet — one per participant**

Participant #: 1 · Date: Fri 28 Aug 2020 · Facilitator: Kalika Nehme

Screen-recording consent: ☐ yes ☐ no · Recording file #: \_\_\_\_\_

| Question                                                             | Answer                                                                                                                                                                        |
|----------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 Reached for first (free look, no instruction) — what/where?        | WALL                                                                                                                                                                          |
| 2 Floor or sofa touched during free look?                            | <input checked="" type="checkbox"/> yes <input type="checkbox"/> no                                                                                                           |
| 3 Time to first unprompted commit (stopwatch, from Task 1 start)     | ____ m ____ s                                                                                                                                                                 |
| 4 Facilitator prompts needed (bally)                                 | <input checked="" type="checkbox"/> a <input checked="" type="checkbox"/> b <input checked="" type="checkbox"/> c <input type="checkbox"/> d <input type="checkbox"/> e       |
| 5 Twist-to-tune discovered unprompted?                               | <input type="checkbox"/> yes <input type="checkbox"/> no <input type="checkbox"/> after prompt                                                                                |
| 6 Samples tried before first commit (bally)                          | <input checked="" type="checkbox"/> a <input checked="" type="checkbox"/> b <input type="checkbox"/> c <input type="checkbox"/> d <input type="checkbox"/> e                  |
| 7 Scheme save/cycle used in Task 2?                                  | <input type="checkbox"/> saved <input type="checkbox"/> cycled <input checked="" type="checkbox"/> neither                                                                    |
| 8 Lamp/light discovered or used?                                     | <input type="checkbox"/> unprompted <input type="checkbox"/> prompted <input checked="" type="checkbox"/> no                                                                  |
| 9 Confidence in final choice (ask: "1-5?")                           | 1 · 2 · 3 · 4 · 5                                                                                                                                                             |
| 10 "Did the limited options feel helpful or restrictive?" (verbatim) | yes good                                                                                                                                                                      |
| 11 Think-aloud: why was a candidate rejected? (verbatim)             |                                                                                                                                                                               |
| 12 One thing that confused you (verbatim)                            | Simultaneous in combination                                                                                                                                                   |
| <b>Menu vs hold-up (20)</b>                                          |                                                                                                                                                                               |
| 13 Method used first after Task 1 prompt                             | <input type="checkbox"/> hold-up <input checked="" type="checkbox"/> menu                                                                                                     |
| 14 Method that produced the kept choice                              | <input type="checkbox"/> hold-up <input checked="" type="checkbox"/> menu <input type="checkbox"/> both                                                                       |
| 15 Menu: opened unprompted? furniture placed?                        | <input type="checkbox"/> opened <input checked="" type="checkbox"/> prompted · <input type="checkbox"/> placed <input type="checkbox"/> moved <input type="checkbox"/> failed |
| 16 Furniture grab (grip) discovered unprompted?                      | <input type="checkbox"/> unprompted <input checked="" type="checkbox"/> prompted <input type="checkbox"/> no                                                                  |

Scripted prompts (use only after ~20 s stuck; tally each in row 4):

- "The floor and sofa are staying — everything else can change."
- "Try touching something that's staying."
- "Try holding it near a wall."
- "Try twisting your wrist while holding it."
- "Point at the floor and press the trigger."
- "Squeeze the grip to grab the furniture; let go to drop it."

Observer notes (gestures, hesitations, errors, stuck previews, anything broken):

changing the highlight color (frame)  
- touch on floor and press the trigger

**IP1 data collection sheet — one per participant**

Participant #: 2 · Date: Fri 28 Aug 2020 · Facilitator: Kalika Nehme

Screen-recording consent: ☐ yes ☐ no · Recording file #: \_\_\_\_\_

| Question                                                             | Answer                                                                                                                                                                        |
|----------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 Reached for first (free look, no instruction) — what/where?        |                                                                                                                                                                               |
| 2 Floor or sofa touched during free look?                            | <input checked="" type="checkbox"/> yes <input type="checkbox"/> no                                                                                                           |
| 3 Time to first unprompted commit (stopwatch, from Task 1 start)     | 3 m ____ s                                                                                                                                                                    |
| 4 Facilitator prompts needed (bally)                                 | <input checked="" type="checkbox"/> a <input checked="" type="checkbox"/> b <input checked="" type="checkbox"/> c <input type="checkbox"/> d <input type="checkbox"/> e       |
| 5 Twist-to-tune discovered unprompted?                               | <input type="checkbox"/> yes <input type="checkbox"/> no <input type="checkbox"/> after prompt                                                                                |
| 6 Samples tried before first commit (bally)                          | <input checked="" type="checkbox"/> a <input checked="" type="checkbox"/> b <input type="checkbox"/> c <input type="checkbox"/> d <input type="checkbox"/> e                  |
| 7 Scheme save/cycle used in Task 2?                                  | <input type="checkbox"/> saved <input type="checkbox"/> cycled <input type="checkbox"/> neither                                                                               |
| 8 Lamp/light discovered or used?                                     | <input type="checkbox"/> unprompted <input type="checkbox"/> prompted <input checked="" type="checkbox"/> no                                                                  |
| 9 Confidence in final choice (ask: "1-5?")                           | 1 · 2 · 3 · 4 · 5                                                                                                                                                             |
| 10 "Did the limited options feel helpful or restrictive?" (verbatim) | Enough for now                                                                                                                                                                |
| 11 Think-aloud: why was a candidate rejected? (verbatim)             |                                                                                                                                                                               |
| 12 One thing that confused you (verbatim)                            | Drop and drop (hand)                                                                                                                                                          |
| <b>Menu vs hold-up (20)</b>                                          |                                                                                                                                                                               |
| 13 Method used first after Task 1 prompt                             | <input type="checkbox"/> hold-up <input checked="" type="checkbox"/> menu                                                                                                     |
| 14 Method that produced the kept choice                              | <input type="checkbox"/> hold-up <input checked="" type="checkbox"/> menu <input type="checkbox"/> both                                                                       |
| 15 Menu: opened unprompted? furniture placed?                        | <input type="checkbox"/> opened <input checked="" type="checkbox"/> prompted · <input type="checkbox"/> placed <input type="checkbox"/> moved <input type="checkbox"/> failed |
| 16 Furniture grab (grip) discovered unprompted?                      | <input type="checkbox"/> unprompted <input checked="" type="checkbox"/> prompted <input type="checkbox"/> no                                                                  |

Scripted prompts (use only after ~20 s stuck; tally each in row 4):

- "The floor and sofa are staying — everything else can change."
- "Try touching something that's staying."
- "Try holding it near a wall."
- "Try twisting your wrist while holding it."
- "Point at the floor and press the trigger."
- "Squeeze the grip to grab the furniture; let go to drop it."

Observer notes (gestures, hesitations, errors, stuck previews, anything broken):

- touch to rotate frame  
- touch just below the frame for "Place it", not drop

P1 — studio, Fri 28 Aug

P2 — studio, Fri 28 Aug

**IP1 data collection sheet — one per participant**

Participant #: 3 · Date: Fri 28 Aug 2020 · Facilitator: Kalika Nehme

Screen-recording consent: ☐ yes ☒ no · Recording file #: \_\_\_\_\_

| Question                                                             | Answer                                                                                                                                                                        |
|----------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 Reached for first (free look, no instruction) — what/where?        | Frame                                                                                                                                                                         |
| 2 Floor or sofa touched during free look?                            | <input checked="" type="checkbox"/> yes <input type="checkbox"/> no                                                                                                           |
| 3 Time to first unprompted commit (stopwatch, from Task 1 start)     | ____ m ____ s                                                                                                                                                                 |
| 4 Facilitator prompts needed (bally)                                 | <input type="checkbox"/> a <input checked="" type="checkbox"/> b <input type="checkbox"/> c <input type="checkbox"/> d <input type="checkbox"/> e                             |
| 5 Twist-to-tune discovered unprompted?                               | <input type="checkbox"/> yes <input type="checkbox"/> no <input type="checkbox"/> after prompt                                                                                |
| 6 Samples tried before first commit (bally)                          | <input checked="" type="checkbox"/> a <input checked="" type="checkbox"/> b <input type="checkbox"/> c <input type="checkbox"/> d <input type="checkbox"/> e                  |
| 7 Scheme save/cycle used in Task 2?                                  | <input type="checkbox"/> saved <input type="checkbox"/> cycled <input checked="" type="checkbox"/> neither                                                                    |
| 8 Lamp/light discovered or used?                                     | <input type="checkbox"/> unprompted <input type="checkbox"/> prompted <input checked="" type="checkbox"/> no                                                                  |
| 9 Confidence in final choice (ask: "1-5?")                           | 1 · 2 · 3 · 4 · 5                                                                                                                                                             |
| 10 "Did the limited options feel helpful or restrictive?" (verbatim) | Feels a bit restrictive                                                                                                                                                       |
| 11 Think-aloud: why was a candidate rejected? (verbatim)             | The walls highlight                                                                                                                                                           |
| 12 One thing that confused you (verbatim)                            | The controller                                                                                                                                                                |
| <b>Menu vs hold-up (20)</b>                                          |                                                                                                                                                                               |
| 13 Method used first after Task 1 prompt                             | <input type="checkbox"/> hold-up <input checked="" type="checkbox"/> menu                                                                                                     |
| 14 Method that produced the kept choice                              | <input type="checkbox"/> hold-up <input checked="" type="checkbox"/> menu <input type="checkbox"/> both                                                                       |
| 15 Menu: opened unprompted? furniture placed?                        | <input type="checkbox"/> opened <input type="checkbox"/> prompted · <input checked="" type="checkbox"/> placed <input type="checkbox"/> moved <input type="checkbox"/> failed |
| 16 Furniture grab (grip) discovered unprompted?                      | <input type="checkbox"/> unprompted <input checked="" type="checkbox"/> prompted <input type="checkbox"/> no                                                                  |

Scripted prompts (use only after ~20 s stuck; tally each in row 4):

- "The floor and sofa are staying — everything else can change."
- "Try touching something that's staying."
- "Try holding it near a wall."
- "Try twisting your wrist while holding it."
- "Point at the floor and press the trigger."
- "Squeeze the grip to grab the furniture; let go to drop it."

Observer notes (gestures, hesitations, errors, stuck previews, anything broken):

change the highlight for a framed one

**IP1 data collection sheet — one per participant**

Participant #: 4 · Date: Fri 28 Aug 2020 · Facilitator: Kalika Nehme

Screen-recording consent: ☐ yes ☒ no · Recording file #: \_\_\_\_\_

| Question                                                             | Answer                                                                                                                                                                        |
|----------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 Reached for first (free look, no instruction) — what/where?        | Sofa                                                                                                                                                                          |
| 2 Floor or sofa touched during free look?                            | <input checked="" type="checkbox"/> yes <input type="checkbox"/> no                                                                                                           |
| 3 Time to first unprompted commit (stopwatch, from Task 1 start)     | ____ m ____ s                                                                                                                                                                 |
| 4 Facilitator prompts needed (bally)                                 | <input checked="" type="checkbox"/> a <input type="checkbox"/> b <input type="checkbox"/> c <input type="checkbox"/> d <input type="checkbox"/> e                             |
| 5 Twist-to-tune discovered unprompted?                               | <input type="checkbox"/> yes <input checked="" type="checkbox"/> no <input type="checkbox"/> after prompt                                                                     |
| 6 Samples tried before first commit (bally)                          | <input checked="" type="checkbox"/> a <input checked="" type="checkbox"/> b <input type="checkbox"/> c <input type="checkbox"/> d <input type="checkbox"/> e                  |
| 7 Scheme save/cycle used in Task 2?                                  | <input type="checkbox"/> saved <input type="checkbox"/> cycled <input checked="" type="checkbox"/> neither                                                                    |
| 8 Lamp/light discovered or used?                                     | <input checked="" type="checkbox"/> unprompted <input type="checkbox"/> prompted <input type="checkbox"/> no                                                                  |
| 9 Confidence in final choice (ask: "1-5?")                           | 1 · 2 · 3 · 4 · 5                                                                                                                                                             |
| 10 "Did the limited options feel helpful or restrictive?" (verbatim) | Feels good for now                                                                                                                                                            |
| 11 Think-aloud: why was a candidate rejected? (verbatim)             | The furniture was not clear                                                                                                                                                   |
| 12 One thing that confused you (verbatim)                            | The controller                                                                                                                                                                |
| <b>Menu vs hold-up (20)</b>                                          |                                                                                                                                                                               |
| 13 Method used first after Task 1 prompt                             | <input type="checkbox"/> hold-up <input checked="" type="checkbox"/> menu                                                                                                     |
| 14 Method that produced the kept choice                              | <input type="checkbox"/> hold-up <input checked="" type="checkbox"/> menu <input type="checkbox"/> both                                                                       |
| 15 Menu: opened unprompted? furniture placed?                        | <input type="checkbox"/> opened <input checked="" type="checkbox"/> prompted · <input type="checkbox"/> placed <input type="checkbox"/> moved <input type="checkbox"/> failed |
| 16 Furniture grab (grip) discovered unprompted?                      | <input checked="" type="checkbox"/> unprompted <input type="checkbox"/> prompted <input type="checkbox"/> no                                                                  |

Scripted prompts (use only after ~20 s stuck; tally each in row 4):

- "The floor and sofa are staying — everything else can change."
- "Try touching something that's staying."
- "Try holding it near a wall."
- "Try twisting your wrist while holding it."
- "Point at the floor and press the trigger."
- "Squeeze the grip to grab the furniture; let go to drop it."

Observer notes (gestures, hesitations, errors, stuck previews, anything broken):

When we select a furniture we should place it, not drop in front

P3 — supplementary, at home (pre-printed date ignored)

P4 — supplementary, at home

IP1 data collection sheet -- one per participant

Participant #: 5 · Date: Fri 28 Aug 2026 · Facilitator: Kalika Nehme

Screen-recording consent: ☐ yes ☒ no · Recording file #: \_\_\_\_\_

|    |                                                                   |                                                                                                                                                                                                                               |
|----|-------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1  | Reached for first (free look, no instruction) -- what/where?      | <u>Wall</u>                                                                                                                                                                                                                   |
| 2  | Floor or sofa touched during free look?                           | <input checked="" type="checkbox"/> yes <input type="checkbox"/> no                                                                                                                                                           |
| 3  | Time to first unprompted commit (stopwatch, from Task 1 start)    | <u>1</u> m <u>    </u> s                                                                                                                                                                                                      |
| 4  | Facilitator prompts needed ( tally)                               | <input checked="" type="checkbox"/> <input checked="" type="checkbox"/> <input checked="" type="checkbox"/> <input type="checkbox"/> <input type="checkbox"/> <input type="checkbox"/> <input type="checkbox"/>               |
| 5  | Twist-to-tune discovered unprompted?                              | <input type="checkbox"/> yes <input checked="" type="checkbox"/> no <input type="checkbox"/> after prompt                                                                                                                     |
| 6  | Samples tried before first commit ( tally)                        | <input checked="" type="checkbox"/> <input checked="" type="checkbox"/> <input type="checkbox"/> <input type="checkbox"/> <input type="checkbox"/> <input type="checkbox"/> <input type="checkbox"/> <input type="checkbox"/> |
| 7  | Schema save/cycle used in Task 2?                                 | <input type="checkbox"/> saved <input type="checkbox"/> cycled <input checked="" type="checkbox"/> neither                                                                                                                    |
| 8  | Lamp/light discovered or used?                                    | <input checked="" type="checkbox"/> unprompted <input type="checkbox"/> prompted <input type="checkbox"/> no                                                                                                                  |
| 9  | Confidence in final choice (ask: "1-5?")                          | 1 <input checked="" type="checkbox"/> 3 <input type="checkbox"/> 4 <input type="checkbox"/> 5                                                                                                                                 |
| 10 | "Did the limited options feel helpful or restrictive?" (verbatim) | <u>red on button station</u>                                                                                                                                                                                                  |
| 11 | Think-aloud: why was a candidate rejected? (verbatim)             | <u>limited selection</u>                                                                                                                                                                                                      |
| 12 | One thing that confused you (verbatim)                            | <u>controls</u>                                                                                                                                                                                                               |
| 13 | Method used first after Task 1 prompt                             | <input type="checkbox"/> hold-up <input checked="" type="checkbox"/> menu                                                                                                                                                     |
| 14 | Method that produced the kept choice                              | <input type="checkbox"/> hold-up <input checked="" type="checkbox"/> menu <input type="checkbox"/> both                                                                                                                       |
| 15 | Menu: opened unprompted? furniture placed?                        | <input type="checkbox"/> opened <input type="checkbox"/> prompted <input checked="" type="checkbox"/> placed <input type="checkbox"/> moved <input type="checkbox"/> failed                                                   |
| 16 | Furniture grab (grip) discovered unprompted?                      | <input type="checkbox"/> unprompted <input checked="" type="checkbox"/> prompted <input type="checkbox"/> no                                                                                                                  |

Scripted prompts (use only after ~20 s elapse; tally each in row 6):

1. "The floor and sofa are staying -- everything else can change."

2. "Try touching something that's staying."

3. "Try holding it near a wall."

4. "Try twisting your wrist while holding it."

5. "Point at the floor and press the trigger."

6. "Squeeze the grip to grab the furniture; let go to drop it."

Observer notes (gestures, hesitations, errors, stuck previews, anything broken):

hand over screen on furniture and maybe

a room check at the end